

Computer Security Summary

Chapter 9:

Hacking Network Devices

This chapter is all about the common devices that make up a network (like routers and firewalls) and how hackers try to break into them.

1. Proxy Servers: The Middleman

- What it is: A server that acts as a gateway between you and the internet. It's like a secretary for your office's internet connection.
- What it does:
 - *Hides your identify*(your computer's IP address) from websites.
 - *Blocks access* to certain websites.
 - *Keeps a log* (a diary) of everyone's internet activity.

- How it's attacked:

- * Hackers can attack the server itself to crash it.
- * They can use it to hide their own identity while attacking other places. To avoid being caught in the proxy's log, they might use a chain of multiple proxies.

2. Routers & Switches: The Traffic Directors

* What they are: Devices that direct data traffic within your network and to the internet. Think of them as “internet traffic cops”.

* How they're attacked:

- * If a hacker gets physical access to the router's console port, they can take full control.
- * Many routers have “default passwords”(like "admin/admin") that people forget to change. Hackers try these first.
- * Hackers can perform “Denial-of-Service (DoS)”attacks to overwhelm and crash them.
- * They can try to change the router's "map" of the network (the routing table) to send data to the wrong place.

3. Firewalls: The Security Guard

*What it is: A device (or software) that acts as a ****bouncer**** for your network, deciding what traffic is allowed in or out based on rules.

- Its Limitations (Important!): Firewalls are not magical force fields.
 - * They can't stop attacks that use “valid usernames and passwords”.
 - * They can't check if the data inside the traffic is malicious.
 - * They only check traffic that goes through them.
 - * If they are not configured correctly, they are useless.
- How they're attacked:
 - *Spoofing: Faking a trusted IP address to trick the firewall.
 - *Back Doors: After hacking a computer inside the network, a hacker creates a secret passage to get back in later, bypassing the firewall.
 - *Root Access: Hackers try to gain full administrative control to change the firewall's rules.

*Through the Web:Since firewalls must allow web traffic, hackers find clever ways to hide attacks within normal-looking web data.

4. VPNs (Virtual Private Networks): The Secure Tunnel

*What it is: A secure, encrypted "tunnel" through the internet that lets employees access their office network from home or a coffee shop.

*How it's attacked: The VPN itself is usually very secure. The problem is the "remote user's computer".

* If a hacker infects an employee's home computer with malware, they can "steal the VPN login details".

* Once the hacker has those details, they can get inside the company's network just like a trusted employee.

How to Protect Everything (The Key Takeaway)

Security is about layers, not just one device. To stay safe:

1. Change Default Passwords! (Especially on routers).
2. Keep Software Updated: This patches known holes hackers exploit.
3. Use Antivirus & Personal Firewalls on all computers, especially those that use VPNs.
4. Educate Users: Don't save VPN passwords on your computer and be careful what you click on.
5. Configure Properly: A misconfigured firewall or router is worse than having none at all.

In a nutshell: This chapter teaches you that every device on a network can be a target. Understanding how they work and how they are attacked is the first step in learning how to defend them.

Chapter 10:

What is a Trojan Horse?

Imagine someone gives you a beautiful gift box. You open it excitedly, but inside is not a gift—it's a thief who sneaks into your house. A

Trojan horse is the digital version of that. It's a *malicious program disguised as something harmless or useful* (like a game, a utility, or a software patch) that tricks you into installing it yourself.

*Key Point: Unlike viruses, Trojans “cannot spread by themselves”. They need you to accidentally invite them in.

- How Do You Get a Trojan?

You can get tricked into downloading and installing a Trojan through:

* Email Attachments: The most common method. You get an email with an attached file that looks important.

* Fake Websites or Downloads: Downloading software from unofficial or hacked websites.

* Social Engineering: Being tricked by clever messages into doing something unsafe, like clicking a link.

* Disguised as Updates: Fake pop-ups or emails pretending to be critical security updates for your software.

- What Can a Trojan Do?

Once installed, a Trojan can give a hacker remote control over your computer. It can:

- * Steal your information: Log your keystrokes (to get passwords), access your files, and take screenshots.

- * Turn your computer into a "zombie": The hacker can use your computer to attack other websites without you knowing.

- * Give full control: The hacker can do anything you can do on your computer: delete files, install more software, or restart your system.

- Famous Trojan Examples

- * PC-Write (1986): The first known Trojan. Disguised as a word processor, it would format your hard drive.

- * AIDS Trojan (1989): Came on a diskette claiming to have AIDS information. It encrypted the user's hard drive and demanded money for the password to decrypt it (an early form of *ransomware*).

- * Back Orifice & NetBus (1998-2001): Gave hackers complete remote control over a victim's computer.

- * Pretty Park (1999): One of the first Trojans to spread like a worm by emailing itself to everyone in your address book.

- * BO2K (Back Orifice 2000): A more advanced version of Back Orifice that could create a network of zombie computers.

- How to Protect Yourself (The Most Important Part!)

The best defense is to *never install a Trojan in the first place.*

- 1.Be Suspicious of Email Attachments: *Never open an attachment* you weren't expecting, even if it appears to be from someone you know.
- 2.Download from Official Sources Only: Only get software from the official company website or official app stores.
- 3.Don't Trust "Too-Good-To-Be-True" Offers: Free games, pirated software, and "exclusive" videos are common ways to hide Trojans.
- 4.Use a Firewall: A firewall can block a Trojan from communicating with the hacker.
- 5.Use Anti-Virus/Anti-Malware Software: Keep it updated so it can recognize and block known Trojans.
- 6.User Accounts: Don't use an "Administrator" account for everyday tasks. Use a standard user account, which can prevent programs from being installed without your password.

Chapter 11:

Denial-of-Service (DoS) Attacks: The Simple Breakdown

Imagine a shop. A *Denial-of-Service (DoS) attack* is like someone (a hacker) blocking the shop's door so that real customers can't get in. The shop's service is "denied" to legitimate users.

A *Distributed Denial-of-Service (DDoS) attack* is even worse. It's like the hacker hiring a huge crowd of people to all block the door at the same time. It's much harder to stop because it's coming from many different places at once.

- Why Do These Attacks Happen? (The Causes)

Hackers can do this because computer systems and networks have weaknesses, like:

*Limited Resources: Every system has a limit on things like internet bandwidth, memory, or processing power.

*Software Bugs: Flaws in the software or operating system that hackers can exploit.

*Network Design: Sometimes the way a network is built has inherent vulnerabilities.

- The Types of DoS Attacks

1. Flood Attacks (Overwhelm with Traffic)

The hacker *floods* the target with a massive amount of useless data, using up all its resources so it can't handle real traffic.

Example: Sending thousands of fake "connection requests" to a web server every second until it crashes.

2. Software Attacks (Exploit a Weakness)

The hacker sends a few, specially crafted malicious packets that exploit a bug in the software, causing it to crash or behave strangely.

Famous Example: Ping of Death: Sending an illegally large "ping" packet that the system didn't know how to handle, causing it to crash. (Most systems are now patched against this)

3. Voluntary DoS: Occurs when the sysadmin has allowed the system to perform a variety of services without considering the system's limitations

4. Involuntary DoS: Takes place regardless of preparation and readiness by the sysadmin

- Isolated vs. Distributed Attacks

*Isolated Attack: Comes from *one single computer*. It's easier to stop—you just block that one source.

*Distributed Attack (DDoS): Comes from hundreds or thousands of computers at the same time. These computers are often infected with malware and controlled by the hacker without the owners' knowledge (they are called "zombies" or "bots"). This is much, much harder to stop.

- Famous Examples of Attacks

*TCP SYN Flood: Fakes a ton of half-finished handshakes with a server, using up all its memory.

*SMURF Attack: Tricks many computers into all replying to a single request, overwhelming the target with responses.

*Botnets: A network of "zombie" computers (the botnet) used to launch massive DDoS attacks or send spam.

- How to Prevent & Stop These Attacks

You can't prevent them 100%, but you can make it much harder and minimize the damage.

- Prevention:

*Keep Everything Updated: Always install the latest security patches for your OS and software.

*Use a Firewall: Configure it to block unwanted traffic.

*Turn Off Unneeded Services: If you don't need it, shut it down. Fewer doors for hackers to try.

*Use Filtering: Set up rules on your network routers to limit suspicious traffic.

➤ Mitigation (What to do during an attack):

*Detect Early: Use intrusion detection systems (IDS) to spot unusual traffic spikes.

*Block the Source: Identify the attacking IP addresses and block them (trickier with DDoS).

*Stay Informed: Know about new attack methods so you can defend against them.